ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 6

NATURAL SCIENCES AND TECHNOLOGY

PAPER 1 · PRACTICE B

MARKS: 70 **TIME: 1½ hours**

An end-of-year test on Grade 6 Terms 3 and 4: Energy and Change (electric circuits, conductors and insulators, mains electricity and safety) and Planet Earth and Beyond (the solar system, the movements of the Earth, the Moon, telescopes and exploring space).

Written by Academic Central for practice, following the CAPS curriculum for Grade 6. It is not a Department of Basic Education examination paper.

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Number your answers exactly as the questions are numbered.
3. Read each question carefully before you answer.
4. Use the correct symbols if you are asked about circuit diagrams.
5. Write neatly and in full sentences where you are asked to explain.

QUESTION 1: MULTIPLE CHOICE

Choose the correct answer. Write only the letter (A–D) next to the question number, e.g. 1.11 B.

1.1 Which material is an insulator?

A Copper B Rubber C Aluminium D Iron

1.2 A circuit with a gap in it is called …

A a closed circuit. B an open circuit.
C a complete circuit. D a power line.

1.3 In a circuit diagram, a long thin line next to a short thick line stands for …

A a bulb. B a cell. C a switch. D a wire.

1.4 A second bulb is added next to the first one in a circuit with one cell. The bulbs will …

A shine more brightly. B shine less brightly.
C shine the same. D blow up.

1.5 Which of these energy sources is renewable?

A Coal B Sunlight C Oil D Natural gas

1.6 The Sun is a …

A planet. B star. C moon. D comet.

1.7 The Earth takes about … to move once around the Sun.

A 24 hours B 28 days C 365¼ days D 7 days

1.8 Which planet is closest in size to the Earth and is called the Earth's "twin"?

A Venus B Mercury C Neptune D Jupiter

1.9 SALT, one of the biggest telescopes in the world, is in the Karoo near …

A Sutherland. B Durban. C Polokwane. D Kimberley.

1.10 The first person to walk on the Moon, in 1969, was …

A Neil Armstrong. B Mark Shuttleworth.
C Yuri Gagarin. D Isaac Newton.

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QUESTION 2: TRUE OR FALSE

Write TRUE or FALSE next to the question number.

2.1 Electricity can only flow in a complete (closed) circuit.

2.2 Plastic is a good conductor of electricity.

2.3 A switch is used to open or close a circuit.

2.4 It is safe to use electrical appliances near water.

2.5 All the planets move around the Sun.

2.6 Mars is further from the Sun than the Earth is.

2.7 The Moon makes its own light.

2.8 A rover is a robot vehicle that can drive on another planet.

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QUESTION 3: ELECTRIC CIRCUITS

FIGURE 1 shows three circuits, P, Q and R. Each circuit has ONE cell.

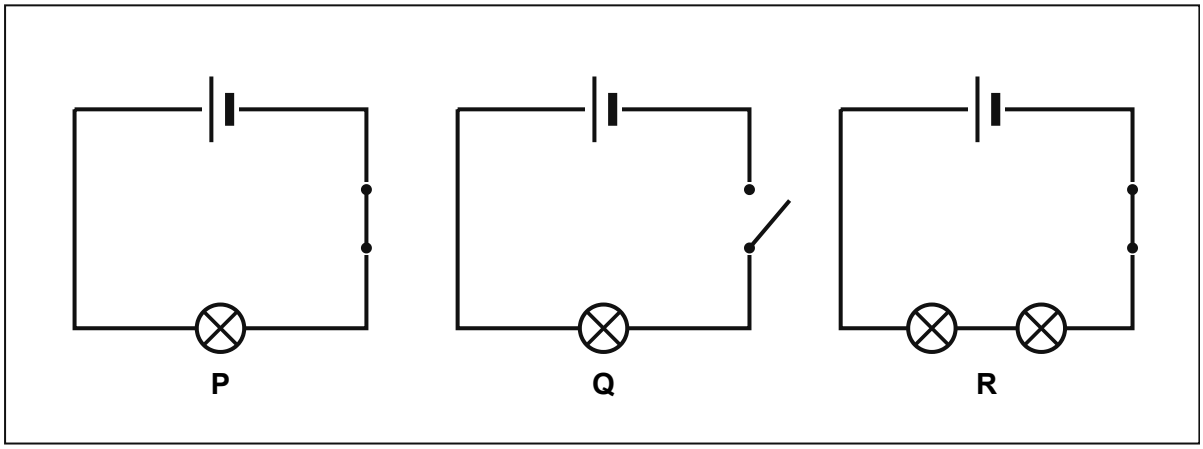


FIGURE 1: Three circuit diagrams

3.1 Name the THREE components (parts) in circuit P, not counting the wires. (3)

3.2 In which circuit will the bulb NOT light up? Give a reason for your answer. (2)

3.3 Will the bulbs in circuit R shine more brightly or less brightly than the bulb in circuit P? Explain why. (2)

3.4 What can you add to circuit R to make its bulbs shine as brightly as the bulb in P? (1)

3.5 What is the job of the cell in a circuit? (1)

3.6 Explain why both ends of the cell must be connected for the bulb to light up. (2)

3.7 Jabu tested different objects in a circuit. His results are in the table.

Object	Did the bulb light up?
copper wire	Yes
wooden pencil	No

aluminium foil	Yes
rubber band	No
glass marble	No
steel paper clip	Yes

3.7.1 Write down the THREE objects that are conductors. (3)

3.7.2 What kind of material are all the conductors made of? (1)

3.7.3 Why do electricians use screwdrivers with plastic or rubber handles? (1)

3.8 When a bulb is switched on, the electrical energy is changed into other forms of energy. Name TWO of them. (2)

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QUESTION 4: MAINS ELECTRICITY AND SAFETY

4.1 Copy and complete the paragraph about a coal power station. Write only the missing words next to the numbers. (3)

> Burning coal changes stored energy into **(4.1.1)** _____ energy. This heat boils water to make steam. The steam turns a turbine, which gives **(4.1.2)** _____ energy. A generator then changes this into **(4.1.3)** _____ energy.

4.2 Which company supplies most of South Africa's electricity? (1)

4.3 Give ONE advantage and ONE disadvantage of using solar panels to make electricity. (2)

4.4 Look at FIGURE 2. Explain why this is dangerous. (2)

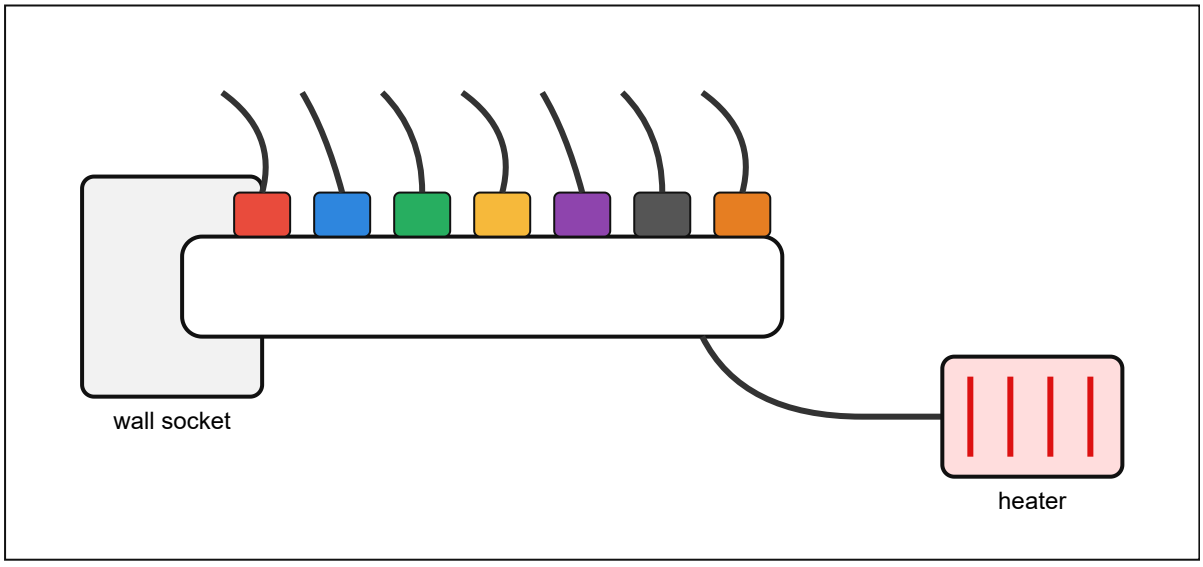


FIGURE 2: Plugs in a multi-plug

4.5 Give TWO safety rules about playing near power lines and pylons. (2)

4.6 You see a person who is getting an electric shock from a broken appliance. Give TWO things you should do. (2)

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QUESTION 5: PLANET EARTH AND BEYOND

5.1 The table gives information about four planets. Study it and answer the questions.

Planet	Distance from the Sun (million km)	Time to move once around the Sun
Mercury	58	88 days
Earth	150	365 days
Mars	228	687 days
Jupiter	778	about 12 years

5.1.1 Which planet takes the shortest time to move around the Sun? (1)

5.1.2 What pattern do you see between a planet's distance from the Sun and the time it takes to move around the Sun? (2)

5.1.3 How many Earth years does Jupiter take to move once around the Sun? (1)

5.1.4 Why is Mercury very hot during the day? (1)

5.2 The Sun seems to rise in the east and move across the sky to set in the west. Explain why it seems to move. (2)

5.3 Answer these questions about the Moon.

5.3.1 What made the round holes (craters) on the Moon? (1)

5.3.2 Why does the full Moon look so bright at night? (1)

5.4 Give ONE difference between a planet and a star. (1)

5.5 Why are big telescopes built on mountains, far away from cities? Give TWO reasons. (2)

5.6 MeerKAT in the Northern Cape is a radio telescope. Why is it built far away from towns and cellphone towers? (1)

5.7 Name TWO things a rover on Mars must be able to do. (2)

5.8 Why will a trip to Mars take much longer than a trip to the Moon? (1)

5.9 Give TWO problems that astronauts have when they live in space for a long time. (2)

5.10 Give TWO ways that satellites moving around the Earth help us in everyday life. (2)

5.11 What is a constellation? Name ONE constellation you can see from South Africa. (2)

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TOTAL: 70

END OF QUESTION PAPER